Lecture 2
Spreadsheet Modeling and the Basics of Managing Data

MGT 207

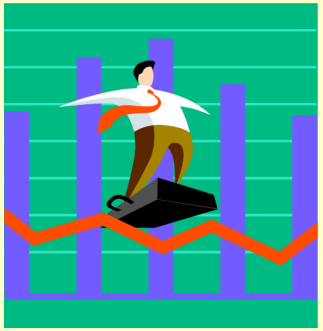
Today's Plan

- Basics of spreadsheet modeling
  - Algebraic modeling, sketch of the spreadsheet
  - Spreadsheet modeling
- Base-case analysis
- "What if" analysis
  - Scenario analysis
  - Data Tables
- Break-even analysis
  - Goal Seek
- Optimization
  - Solver

Common Use of Spreadsheets

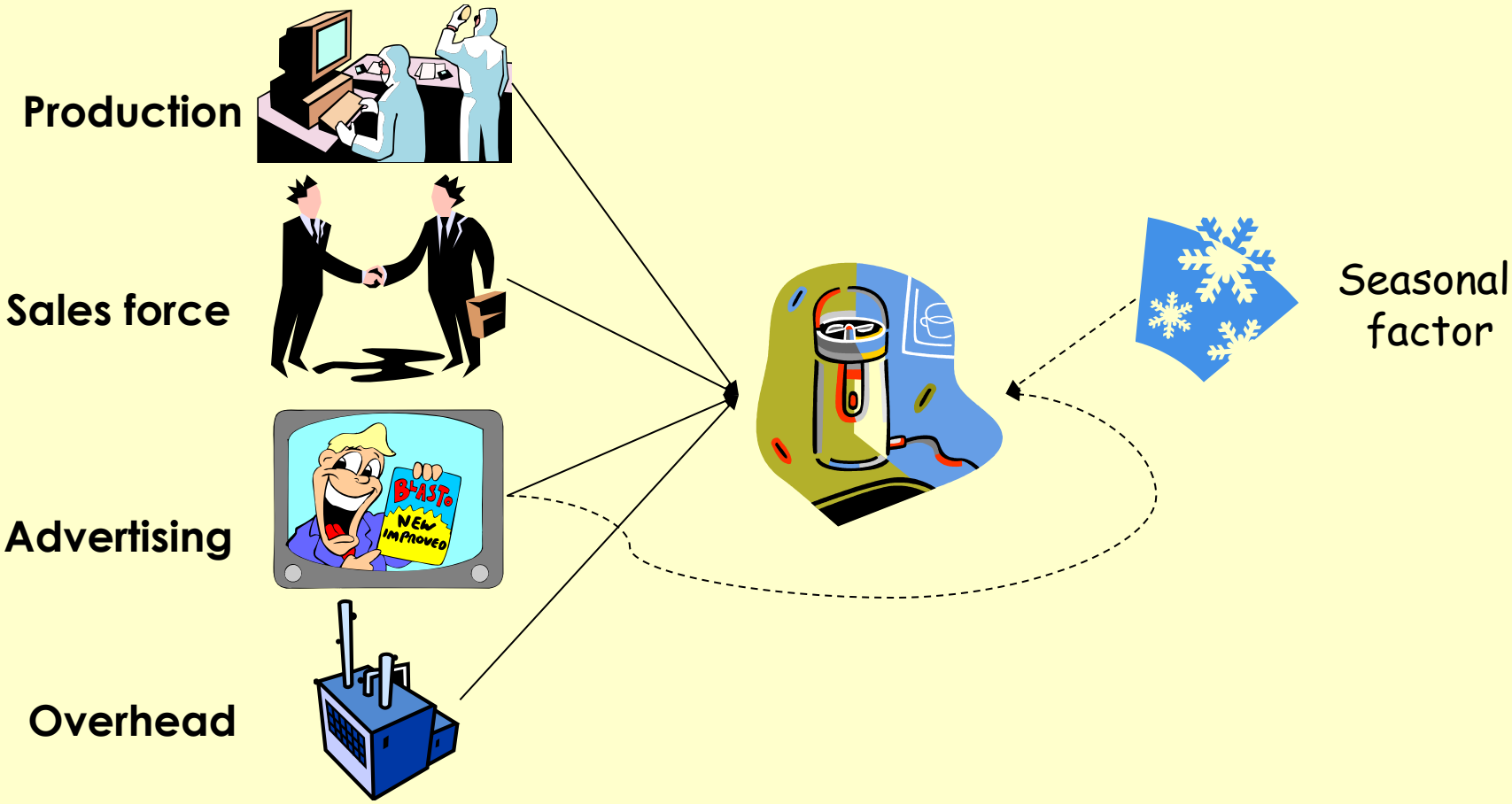
- The spreadsheet as a calculator
  - easy access to the standard mathematical functions, probability distributions
  - easy "what-if" calculations
- **The spreadsheet as a mathematical tool**
  - **simulation**
  - **optimization**

- The spreadsheet as a statistical tool
  - regression
  - graphical analysis
- The spreadsheet as a database
  - pivot tables



Example: Advertising Budget Planning

- How do we allocate the advertising budget to maximize profit?
- How do various factors affect our profitability ?



Algebraic Model: Question 1

- Decision variables: **Q1, Q2, Q3, Q4 advertising budget**
- Structure of the model:

Profit = Revenue - **Total Cost**

Revenue = Price * Unit sales

Total Cost

= production cost + sales expenses+ advertising +overhead

= unit prod. cost * **units sold** + sales expenses (fixed)

+ advertising (decision variable) +15%*revenue

Units sold = 35*Seasonal factor * SQRT(3000+Advertising)

(1)Where does the formula come from? (2) Is it a problem that the result is not an integer? (3) Is the SQRT() function in the formula reasonable? That is, could we for example have:

Units sold=Seas. fact * Exp(Advertising)

Sketch of the Spreadsheet

Parameters

DECISIONS	Q1	Q2	Q3	Q4
Advertising				

OUTPUTS
Profits

CALCULATIONS

	Q1	Q2	Q3	Q4
Units Sold	=Use Advertising and sales formula			
Revenue	=Price*Units Sold			
Cost of Goods	=Production Cost*Units Sold			
Gross Margin	=Revenue – Cost of Goods			
Sales Expenses	=fixed at 8000 or 9000			
Advertising	=the decision variables			
Overhead	=fixed fraction of Revenue			
Total of other costs	=Sales Expenses + Advertising + Overhead			
Profit	=Gross Margin – Total of other costs			

Spreadsheet Model – Calculation for Q1

	A	B	C	D	E	F	G	H
1	Adervertising Budget Model							
2	LD							
5	PARAMETERS							
6				Q1	Q2	Q3	Q4	
7			Price	$40				
8			Production Cost	$25				
9			Seasonal	0.9	1.1	0.8	1.2	
10			Overhead rate	0.15				
11			Sales parameters					
12				35				
13				3000				
14			Sales Expense	8000	8000	9000	9000	
15			Ad Budget	$40,000				
17	DECISIONS							Total
18		Ad Expenditures		10000	10000	10000	10000	40000
20	OUTPUT							
21		Profit			Base case			
23	CALCULATIONS							Total
24		Quarter		Q1	Q2	Q3	Q4	
25		Seasonal factor		0.9	1.1	0.8	1.2	
26								
27		Units Sold						
28		Revenue						
29		Cost of Goods						
30		Gross Margin						
31								
32		Sales Expense						
33		Advertising						
34		Overhead						
35		Total Other Cost						
36								
37		Profit						
38		Profit Margin						

Inputs

Decisions

Outputs

Calculation

Cell	Formula
D27	
D28	
D29	
D30	
D32	
D33	
D34	
D35	
D37	
D38	

Spreadsheet Model – Calculation for Q1

	A	B	C	D	E	F	G	H
1	Adervertising Budget Model							
2	LD							
5	PARAMETERS							
6				Q1	Q2	Q3	Q4	
7			Price	$40				
8			Production Cost	$25				
9			Seasonal	0.9	1.1	0.8	1.2	
10			Overhead rate	0.15				
11			Sales parameters					
12				35				
13				3000				
14			Sales Expense	8000	8000	9000	9000	
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17	DECISIONS							Total
18		Ad Expenditures		10000	10000	10000	10000	40000
20	OUTPUT							
21		Profit			Base case			
23	CALCULATIONS							Total
24		Quarter		Q1	Q2	Q3	Q4	
25		Seasonal factor		0.9	1.1	0.8	1.2	
26								
27		Units Sold						
28		Revenue						
29		Cost of Goods						
30		Gross Margin						
31								
32		Sales Expense						
33		Advertising						
34		Overhead						
35		Total Other Cost						
36								
37		Profit						
38		Profit Margin						

Inputs

Decisions

Outputs

Calculation

Cell	Formula
D27	=**D12***D25 *sqrt(**D13**+D18)
D28	=**D7***D27
D29	=**D8***D27
D30	=D28-D29
D32	=D14
D33	=D18
D34	=**D10***D28
D35	=SUM(D32:D34)
D37	=D30-D35
D38	=D37/D28

Copy cells **D27:D38**
To
E27:G38

Spreadsheet Model – Base Case (Benchmark)

	A	B	C	D	E	F	G	H
1	Adervertising Budget Model							
2	LD							
5	PARAMETERS							
6				Q1	Q2	Q3	Q4	
7			Price	$40				
8			Production Cost	$25				
9			Seasonal factor	0.9	1.1	0.8	1.2	
10			Overhead rate	0.15				
11			Sales parameters					
12				35				
13				3000				
14			Sales Expense	8000	8000	9000	9000	
15			Ad Budget	$40,000				
17	DECISIONS							Total
18		Ad Expenditures		10000	10000	10000	10000	40000
20	OUTPUT							
21		Profit	69662.10356		Base case	69662.10356		
23	CALCULATIONS							
24		Quarter		Q1	Q2	Q3	Q4	Total
25		Seasonal factor		0.9	1.1	0.8	1.2	
26								
27		Units Sold		3591.552589	4389.675387	3192.49119	4788.736785	15962.46
28		Revenue		$143,662.10	$175,587.02	$127,699.65	$191,549.47	638498.2
29		Cost of Goods		$89,788.81	$109,741.88	$79,812.28	$119,718.42	399061.4
30		Gross Margin		53873.28884	65845.1308	47887.36785	71831.05178	239436.8
31								
32		Sales Expense		8000	8000	9000	9000	34000
33		Advertising		10000	10000	10000	10000	40000
34		Overhead		21549.31553	26338.05232	19154.94714	28732.42071	95774.74
35		Total other Cost		39549.31553	44338.05232	38154.94714	47732.42071	169774.7
36								
37		Profit		14323.9733	21507.07848	9732.420712	24098.63107	69662.1
38		Profit Margin		0.099705997	0.122486725	0.076213372	0.125808914	0.109103

Cell	Formula
H27 =	SUM(D27:G27)

Copy cell **H27**
To **H28:H37**

Copy cell **G38**
To cell **H38**

Cell	Formula
C21	= H37

Let cell **F21** store the Base case:
Copy cell **C21**
Paste Special …
Values to cell
F21

Features in Excel Spreadsheet

- Absolute reference vs. Relative reference:
  - $
  - Function Key F4
- Comments :
  - Insert -> Comment **[03]**
  - Review Tab -> Comments Group -> New Comments **[07]**
- View Formula sheet:
  - Ctrl + `
- Print with row and column headings:
  - File -> Page Setup… ->Sheet -> row and column headings **[03]**
  - Print -> Print Preview -> Page Setup… -> Sheet -> row and column headings **[07]**
- Plots:
  - Chart Wizard **[03]**
  - Insert Tab -> Charts Group **[07]**
- Auditing:
  - Tools -> Formula Auditing **[03]**
  - Formulas Tab -> Formula Auditing **[07]**

Rules of Spreadsheet Engineering

BEST PRACTICE GUIDELINES IN SPREADSHEET DESIGN:

- Sketch the spreadsheet
- Organize the spreadsheet into modules
  - Isolate input parameters
- Keep it simple
  - Avoid long formulas
  - Formulas only contain cell references (i.e. not numbers); this simplifies spreadsheet auditing
- Design for communication
  - Use Excel formatting options (Outlines, color, bold font, and so on) to highlight certain cells for quick visual recognition
- Document important data and formulas

"What if" Analysis – Data Table

Question 2: What if production costs are 10% higher or 10% lower than assumed?

Answer: Modify the corresponding parameter cell (D8)

Extra question: How do changes in Q1 and Q2 ad expense affect our profitability?

Answer: Use **DataTable** feature in Excel

- Data -> Table … **[03]**
- Data Tab -> Data Tools -> What-if Analysis -> Data Table… **[07]**

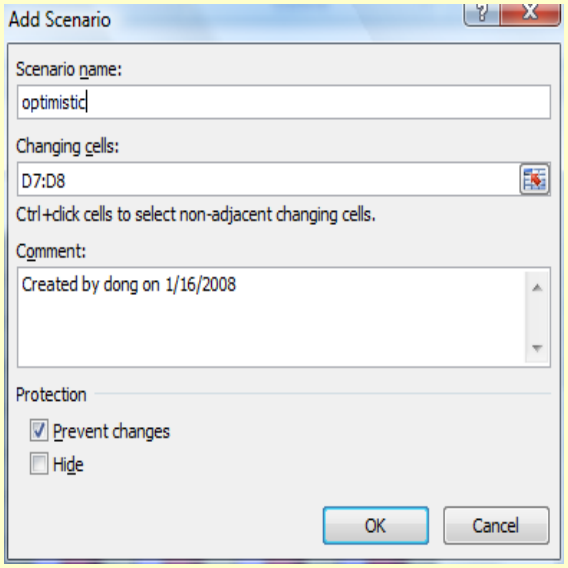
See details in **SensitivityAnalysis.xls**
(Blackboard, Course Outline, Session 2)

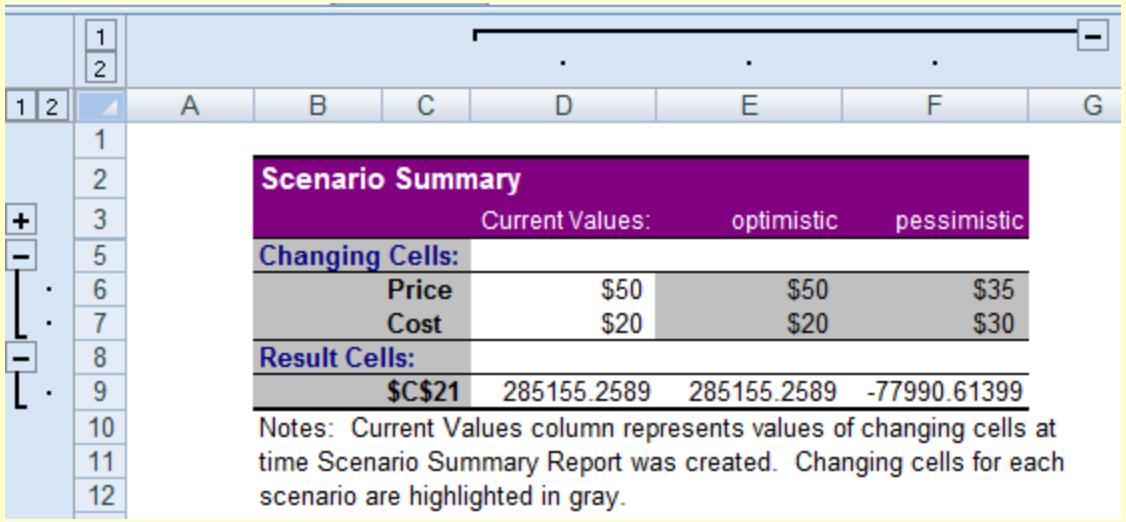
1. Two-way Table
2. One-way Table

- Result interpretation – scatter graph and 3D heatmap

"What if" Analysis – Scenario Analysis

- Scenario: a story about the future decision-making environment translated into its effects on several of the model's parameters, e.g., "business as usual," "Price War," "Market Discipline."
- For example,
  - Optimistic scenario: price=$50, cost=$20
  - Pessimistic scenario: price =$35, cost =$30
- **Tools -> Scenarios (add/edit) [03]**
- **Data Tab -> Data Tools -> What-if Analysis -> Scenario Manager... (07)**





Scenario Summary			
	Current Values:	optimistic	pessimistic
Changing Cells:			
Price	$50	$50	$35
Cost	$20	$20	$30
Result Cells:			
C21	285155.2589	285155.2589	-77990.61399

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

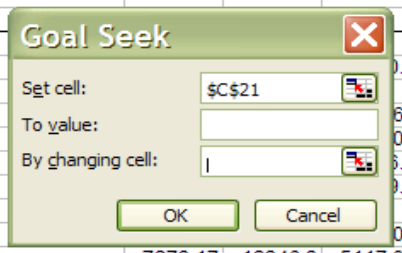
Break-even Analysis - Use of Goal Seek

Question 3: <u>What production cost will bring a zero profit?</u>

Answer: **Tools -> Goal Seek ... (03)**

Data Tab -> Data Tools -> What-if Analysis -> Goal Seek... (07)

	A	B	C	D	E	F	G	H
5	PARAMETERS							
6				Q1	Q2	Q3	Q4	
7			Price	$40				
8			Production Cost	$25				
9			Seasonal factor	0.9	1.1	0.8	1.2	
10			Overhead rate	0.15				
11			Sales parameters					
12				35				
13				3000				
14			Sales Expense	8000	8000	9000	9000	
15			Ad Budget	$40,000				
17	DECISIONS							Total
18		Ad Expenditures		7273.17	12346.3	5117.07	15263.4	40000
20	OUTPUT							
21		Profit	71446.7944		Base cas	69662.1		
23	CALCULATIONS							
24		Quarter					Q4	Total
25		Seasonal factor				.8	1.2	
26								
27		Units Sold				65	5675.97	16160.8
28		Revenue				06	227039	646430
29		Cost of Goods				.4	141899	404019
30		Gross Margin				.8	85139.6	242411
31								
32		Sales Expense				00	9000	34000
33		Advertising		7273.17	12346.3	5117.07	15263.4	40000
34		Overhead		19156.4	28616.4	15135.9	34055.8	96964.5
35		Total Fixed Cost		34429.6	48962.7	29253	58319.2	170965
36								
37		Profit		13461.4	22578.2	8586.82	26820.3	71446.8
38		Profit Margin		0.10541	0.11835	0.0851	0.11813	0.11053



Goal Seek Parameters:

Set cell: C21

the cell containing **profit formula**

To values: 0

for break-even.

Other numbers for different managerial situations.

By change cell: D8

the cell contains **production cost parameters**.

Or cells that are of your interests

Optimization – Use of Solver

Question 4: <u>What is the maximum profit we can make with an advertising budget of $40,000?</u>

Answer: **Tools** -> **Solver …**

Note: Make sure Solver is installed on your computer. Look for it in Add-Ins.

Use of Solver (Cont.)

Question 4: <u>What is the maximum profit with an advertising budget of $40,000?</u>

	A	B	C	D	E	F	G	H
5	PARAMETERS							
6				Q1	Q2	Q3	Q4	
7			Price	$40				
8			Production Cost	$25				
9			Seasonal factor	0.9	1.1	0.8	1.2	
10			Overhead rate	0.15				
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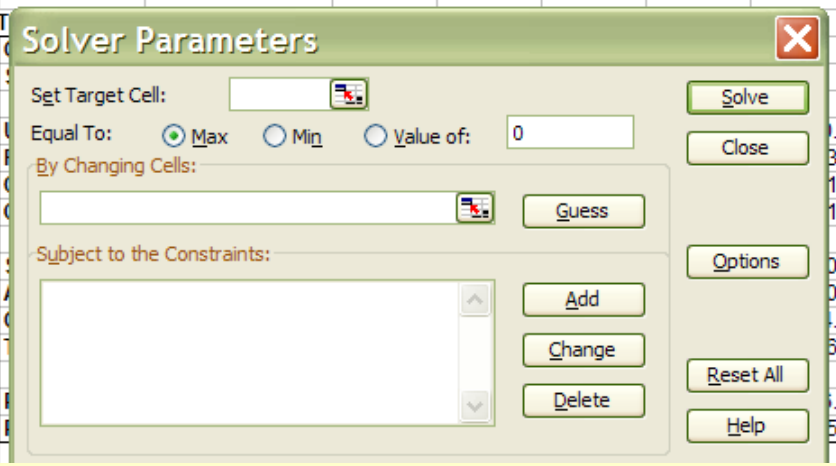
12				35				
13				3000				
14			Sales Expense	8000	8000	9000	9000	
15			Ad Budget	$40,000				
17	DECISIONS							Total
18		Ad Expenditures		7273.17	12346.3	5117.07	15263.4	40000
20	OUTPUT							
21		Profit	71446.7944		Base cas	69662.1		
23	CALCULAT							

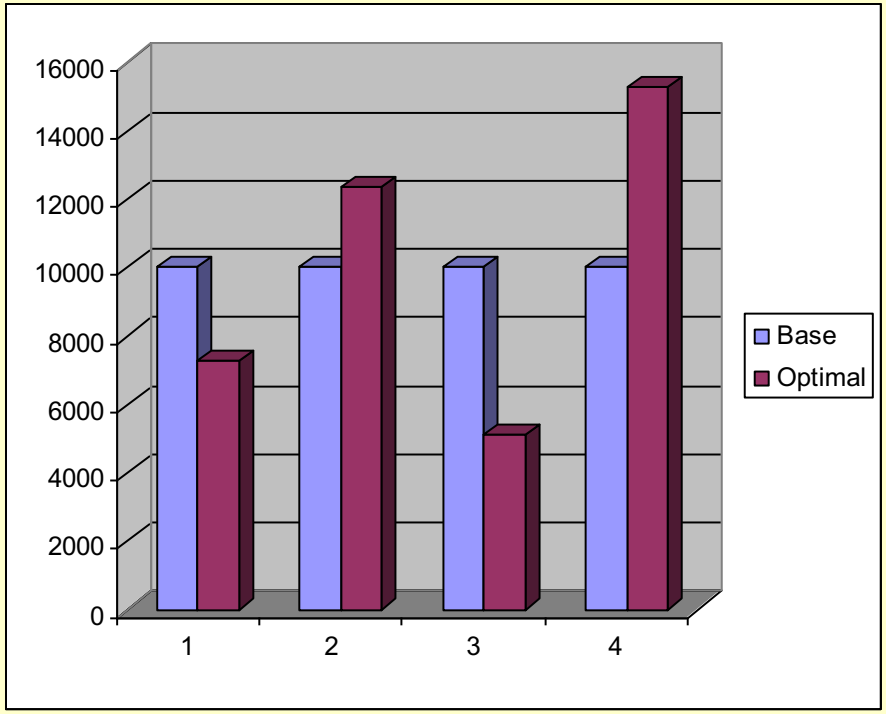


Solver Parameters:
Set Target cell: C21
the cell containing **profit function**

By changing cells: D18:G18
Other cells for different managerial situations.

Subject to the Constraints:
H18≤D15

Comparison of Base Case and Optimal Case



Does the optimal allocation make sense?

What does the result implicitly assume?

Inventory Problem at L.L. Bean, Inc.

Inventory Problem at LL Bean

- **Problem description:**
  - Single item with a single ordering opportunity and one selling season with random demand.
  - Selling price = p = \$45.
  - Cost = c = \$25
  - End of season discount price = v = \$15
  - Demand forecast = 12000 units
  - Demand distribution = U[10000,14000]

- **Task:**
  - Determine LL Bean's optimal stocking quantity